MASTERS OF BUSINESS ADMINISTRATION

SEMESTER 1



DMBA115

MANAGERIAL ECONOMICS

Unit - 6

Price Determination under Perfect & Imperfect Competition

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1. INTRODUCTION

Understanding how prices are determined under various market structures is crucial for business managers to make informed pricing and strategic decisions. Markets can broadly be classified into perfect and imperfect competition, each with distinct characteristics that influence pricing behavior and market dynamics. In a perfectly competitive market, numerous buyers and sellers interact, ensuring no single entity can influence the market price. Products are homogeneous, and there are no barriers to entry or exit, allowing firms to freely enter or leave the market based on profitability. Price determination in such markets is straightforward, governed by the forces of supply and demand. Individual firms are price takers, accepting the market price as given. For example, agricultural markets, where numerous farmers sell identical products like wheat or rice, often exhibit features of perfect competition. Despite its theoretical robustness, perfectly competitive markets are rarely observed in the real world due to the practical limitations of homogeneity and complete information. In contrast, imperfect competition dominates real-world markets due to variations in market characteristics. These markets include structures such as monopolies, monopolistic competition, and oligopolies, where firms possess some degree of market power to influence prices. Imperfect markets arise due to brand differentiation, proprietary technologies, media influence, government regulations, and unequal access to financial resources. For instance, branded goods like luxury watches or technology products operate in imperfect markets, where firms use pricing strategies to differentiate themselves and gain a competitive edge.

This unit examines both perfect and imperfect competition, highlighting how pricing decisions are made in these contrasting environments. By understanding the dynamics of these market structures, managers can develop effective pricing strategies to navigate competition and align with consumer expectations.

1.1 Objectives



After studying this unit, you should be able to:

- Analyse the market with respect to its structure of competition
- Differentiation between different types of market structures
- Explain how firms in perfectly competitive markets maximize their output
- Apply short-run and long-run concepts to price determination in perfectly competitive markets
- Explain how firms operating in imperfectly competitive markets maximize their output
- Formulate realistic estimates of profit maximization.
- Examine how price discrimination can be applied.
- Describe the working of oligopoly in practice

2. MARKET AND MARKET STRUCTURE

Market, in economics, does not refer to a place or places but to a commodity and also to buyers and sellers of that commodity who are in competition with one another, for example, the cotton market may not be confined to a particular place, but may cover the entire country and, in fact, even the entire world. Buyers and sellers of cotton may be spread all over the world.

Thus, in common parlance, market refers to a place where sellers and buyers for exchanging goods, but in the language of economics it has a wider meaning. It refers to a context where buyers and sellers come into close contact with one another for the settlement of their transactions.

According to Prof. Cournot, the term market is, "not any particular market place in which things are bought or sold, but the whole of any region in which buyers and sellers are in such free interaction with one another that the price of the same goods tends to equalize easily and quickly".

In the words of Prof. Benham, Market is, "any area over which buyers and sellers are in such close touch with one another, either directly or through dealers that the prices obtainable in one part of the market affects the prices paid in other parts of the market". For the existence of a market, there is no need for face-to-face contact between the buyers and sellers to conclude their transactions. In recent years, means of transport and communication have developed so fast that buyers and sellers can easily come into close contact with each other for the settlement of their transactions without establishing a face-to-face relationship.

The term market hence implies to:

- Existence of a commodity to be traded.
- Existence of sellers and buyers.
- Establishment of contact between the sellers and buyers.
- Willingness and ability to buy and sell a commodity and
- The existence of a price at which the given commodity is to be bought and sold.

Market situation varies in structure. Market structure refers to economically significant features of a market, which affect the behavior and working of firms in the industry. It tells us how a market is built up and what its basic features are. According to Pappas and Hirschey, "Market structure refers to the number and size distribution of buyers and sellers in the market for a good or service". It indicates a

set of market characteristics that determine the nature of the market in which a firm operates. Different market structures affect the behavior of sellers and buyers in different ways.

The primary characteristics of markets are as follows:

- The number and size distribution of sellers
- The number and size distribution of buyers
- Product differentiation
- Conditions of entry and exit

The number and size distribution of sellers

A market may consist of many, few or very few sellers. There may be a few big firms with huge investments or a large number of small firms with limited investments. Thus, the operating size of the firm may be large or small in a market. The number and size of sellers influence the working of a market.

The number and size distribution of buyers

In a market, there may be a large number of buyers. Similarly, a market may consist of many small buyers or only a few buyers. The total number of buyers influences the nature of transactions in the market.

Product differentiation

Products sold in the market may be homogeneous, or have substitutes, close substitutes or remote substitutes. A firm may deliberately differentiate its product from that of the products of other firms by adopting several techniques.

Conditions of entry and exit

In a few market situations, new firms may enter the industry or old firms may leave the industry of their own free will. In the case of other market situations, there will be deliberate entry barriers.

Thus, the characteristics of market structure give us information about the nature of working of different markets.

Among the different market situations, perfect competition and monopoly form the two extremes. In between these two market situations, we come across a number of market situations which may be collectively termed as imperfect markets. In these imperfect markets, we notice the elements of competition as well as monopoly. They are bi-lateral monopoly, monopsony (one buyer), duopoly

(two sellers) duopsony (two buyers), oligopoly (few sellers), oligopsony (few buyers) and monopolistic competition (many sellers). Figure 1 shows the types of competition.

Market Situation		
Perfect Competition	Imperfect Competition	Monopolistic Competition Bilateral Monopoly Oligopoly Duopoly Monopsony Duopsony Oligopsony

Figure 1: Table showing the Types of Competition

The market situations vary in structure. Different market structures affect the behavior of buyers and sellers and firms. Further, prices and trade volumes are influenced by different types of markets and price - output determination under different market conditions.

3. PERFECT COMPETITION

Perfect competition is a comprehensive term which includes pure competition too. Before we discuss the details of perfect competition, it is necessary to have a clear idea regarding the nature and characteristics of pure competition.

Pure Competition is a part of perfect competition. Competition in the market is said to be pure when the following conditions are satisfied:

- Prevalence of a large number of buyers and sellers.
- The commodity supplied by each firm is homogeneous.
- Free entry and exit of firms.
- Absence of any kind of monopoly element.

Under these conditions, no individual producer is in a position to influence the market price of the product. According to Prof. E.H. Chamberlin – **"Under Pure Competition, as the individual seller's market is completely merged with the general one, he can sell as much as he pleases at the going price"**. Further, he remarks, "Pure competition means unalloyed by monopoly elements. It is a much simpler and less exclusive concept than perfect competition".

Prof. Joel Dean, after going through the features of pure competition, observes that, "Pure competition does exist in reality but it is a rare phenomenon". Hence, it is stated that it is possible to come across pure competition in our life, for example, in the markets for rice, wheat, cotton, jowar, fruits, vegetables, eggs, etc., where there are a large number of sellers and buyers and practically all goods are identical. If we look at the present market, we notice that even in these cases, there is a possibility of cartels forming by sellers to influence the market price. Now, we shall turn our attention to perfect competition.

Meaning and definition of perfect competition

A perfectly competitive market is one in which the number of **buyers and sellers is large, all engaged in buying and selling a homogeneous product without any artificial restriction and possessing perfect knowledge of the market at the same time.** According to Bilas, "the perfect competition is characterized by the presence of many firms; they all sell the same product which is identical. The seller is the price-taker". According to Prof. F. Knight, perfect competition entails

"Rational conduct on the part of buyers and sellers, full knowledge, absence of friction, perfect mobility and perfect divisibility of factors of production and completely static conditions".

Features of perfect competition

1. **Existence of a large number of buyers and sellers**

   A perfectly competitive market will have a large number of sellers and buyers. The output of a seller (firm) will be so small that it is a negligible fraction of the output of the industry. Hence, changes in supply made by a particular firm will not affect the total output or price. Similarly, no single buyer can influence the price of the commodity because the quantity purchased by him is a small fraction of the total quantity.

2. **Homogenous products**

   Different firms constituting the industry produce homogenous goods. They are identical in character. Hence, no firm can raise its price above the general level.

3. **Free entry and exit of firms**

   There is absolute freedom for firms to get in or get out of the industry. If the industry is making profits, new firms are attracted into the industry. Conversely, firms will quit the industry if there are losses. This results in the realization of normal profits by all the firms in the long run.

4. **Existence of single price**

   Each unit bought and sold in the market commands the same price since products are homogeneous.

5. **Perfect knowledge of the market**

   All sellers and buyers will have perfect knowledge of the market. Sellers cannot influence buyers and vice versa.

6. **Perfect mobility of factors of production**

   Factors of production are free to move into any industry or occupation in order to earn higher rewards. Similarly, they are also free to come out of the occupation or industry if they feel that they are under remunerated.

7. **Full and unrestricted competition**

A perfectly competitive market is free from all sorts of monopoly and oligopoly conditions. Since there are a large number of buyers and sellers, it is difficult for them to join together and form cartels or form organizations. Hence, each firm acts independently.

8. **Absence of transport cost**

   All firms will have equal access to the market. Market price charged by the sellers should not vary because of the difference in the cost of transportation.

9. **Absence of artificial government controls**

   The government should not interfere in matters pertaining to supply and price. It should not place any barriers in the way of smooth exchange. The price of a commodity must be determined only by the interaction of supply and demand forces.

10. **The market price is flexible over a period of time**

    Market price changes only because of changes in either demand or supply force or both. Thus, price is not affected by the sellers, buyers, firm, industry or the government.

11. **Normal profit**

    As the market price is equal to the cost of production, firms in perfectly competitive markets can earn only normal profits. Normal profits are those which are just sufficient to ensure the firms stay in business. It is the minimum reasonable level of profit which the entrepreneur must get in the long run. It is a part of the total cost of production because it is the price paid for the services of the entrepreneur, i.e., profit is an item of expenditure for a firm.

Special features of perfect competition

i. It is an extreme form of market situation rarely found in the real world.
ii. It is a mere concept, a myth, an illusion and purely theoretical in nature.
iii. It is a hypothetical model.
iv. It is an ideal market situation.

Reasons for the study of perfect competition

1. It is used as a yardstick against which all other models can be compared and evaluated.
2. It is quite accurate and useful in explaining and predicting the behavior of a market and a firm under certain circumstances.

3. It is a good simplified model for beginners to start with.The study is useful to prepare grounds for future study of imperfect markets.
4. It is a useful model to compare the actual with the ideal; what is and what ought to be.
5. It helps us to understand optimum allocation of resources in an ideal market.

4. PRICE – OUTPUT DETERMINATION UNDER PERFECT COMPETITION (GENERAL MODEL)

Studying the price - output model under perfect competition is quite interesting. In the case of the industry, under a perfectly competitive market, market price of the product is determined by the interaction of supply and demand. The market price is not fixed by the buyer or the seller, firm, industry or by the government. It is only the market forces, i.e., demand and supply, that determine the equilibrium price of the product. This peculiar feature is seen only under perfect competition.

Alfred Marshall compared supply and demand to the blades of a scissor. Just as both the blades work together to cut a piece of cloth, both supply and demand interact with each other to determine the market price at which exchange takes place. In the process of price determination, supply is not more important than demand or, demand is not more important than supply. Both forces play an equally important role.

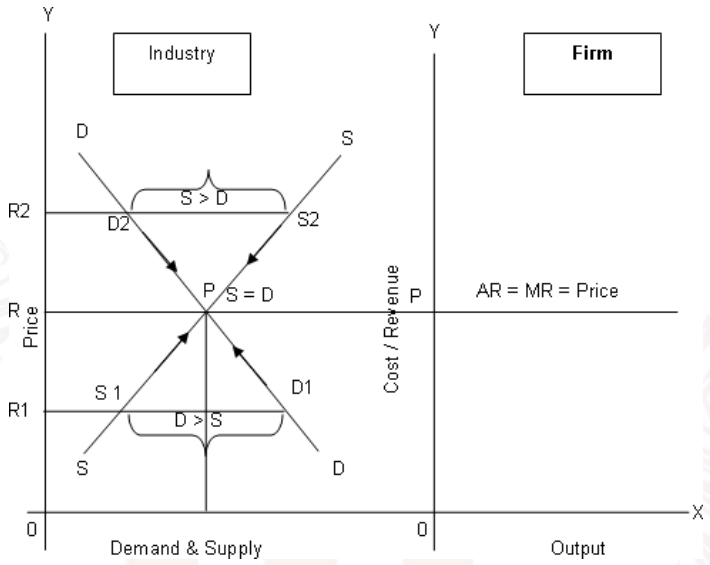
Table 1 depicts how the price is determined in the market by the interaction of demand and supply.

Table 1: Price Determination Based on Demand and Supply

Price in Rs.	Demand in Units	Supply in Units	State of Market	Pressure on Price
10	1000	9000	Surplus S > D	Downward
8	3000	7000	Surplus S > D	Downward
6	5000	5000	Equilibrium S =D	Neutral
4	7000	3000	Shortage D > S	Upward
2	9000	1000	Shortage D > S	Upward

From the table, it is clear that equilibrium price is determined at Rs. 6.00 where demanded quantity is exactly equal to quantity supplied i.e., 5000 units. Figure 2 shows the equilibrium output.

Figure 2: Equilibrium Output



In the case of any industry, interaction of supply and demand will determine the equilibrium market price. In Figure 2, P indicates OR as equilibrium price and OQ as equilibrium output. **The price at which demand and supply are equal is known as equilibrium price. The quantity bought and sold at the equilibrium price is known as equilibrium output.**

In the figure, equilibrium price is determined at point P where both demand and supply are equal. The upper limit of the price of a product/service is determined by the demand. This price should not exceed 'what the market can bear'. In short, the price of the product / service should not exceed the value of its benefit to the buyers (price should not be more than the utility of product / service).

The lower limit of the price is determined by the production cost. In the long run, the price should not fall below the production costs of making and distributing the product / service. With reference to industry, point P can be regarded as the position of stable equilibrium. Even if there are changes in the price, there will be automatic adjustments in supply and demand, restoring the original equilibrium position. When the price rises from OR to OR1 supply exceeds demand, there will be excess supply over demand. The excess supply of goods pushes down the price from OR1 to OR, the original price.

Similarly, when price falls from OR to OR2, demand exceeds supply, excess demand over supply in its turn pushes up the prices from OR2 to OR - the original price. Thus, interaction between demand and supply determines the market price.

Under perfect competition, a firm will not have any independence to fix the price of its own product. The industry is the price - maker or giver and a firm is a price - taker or price acceptor and quantity adjuster. As a part of the industry, it has to simply charge the price which is determined by the industry. If it charges a higher price, it will lose its sales and, if it charges a lesser price, it will incur losses.

In the case of a firm, the price line which is equal to AR and MR, will be horizontal and parallel to OX - axis. This is because the same price has to be charged by the firm for all the units supplied, irrespective of changes in demand. Hence,

Equilibrium or Market Price = AR = MR

Difference between a firm and an industry

Basically, there is a difference between a firm and an industry. **A firm is a single manufacturing unit producing and selling either a commodity or a service.** It is a part of industry and is called a business enterprise. Business is an economic activity and a business unit is an economic unit and an individual production unit. It converts inputs into outputs. These production units are organized and run by people either as individuals or as members of households or as a group of people. **It is basically an income-generating unit.** It buys inputs like raw materials, labor, capital, power, fuel, etc. and produces goods and services for the consumers. It organizes and combines all kinds of resources and plans for the use of these resources in the best possible manner.

Profit making is the basic objective of a firm. The traditional and conventional objective of a firm was profit maximization and now, the main objective is profit optimization.

A business firm is a **legal entity** on the basis of ownership and contractual relationship organized for the production and for the sale of goods and services.

Many firms producing similar or homogeneous goods or services collectively make an industry. The term industry refers to a set or group of firms engaged in the production of a particular product or service. For example, Reid and Taylor, Dignam, Reliance Industries, Raymond Ltd., etc are all firms producing textiles. Such firms put together constitute the textile industry in India. Thus, an

industry is engaged in the production of homogeneous goods that are substitutes for each other, use common raw materials, have similar processes, etc. **All firms engaged in providing the same kind of services or doing a common trade or business constitute an industry,** for example, banks, hotels, etc. An industry is a particular line of productive activity in which many firms are engaged, each adopting its own production and pricing policies to its best advantage.

5. EQUILIBRIUM OF THE INDUSTRY AND THE FIRM UNDER PERFECT COMPETITION

Short-run Industry Equilibrium under Perfect Competition

The term 'Equilibrium' in physical science implies a state of balance or rest. In economics, it refers to a position or situation from which there is no incentive to change. **At the equilibrium point, an economic unit is maximizing its benefits or advantages.** Hence, there will always be a tendency on the part of each economic unit to move towards the equilibrium condition. Reaching the position of equilibrium is a basic objective of all firms.

In the short period, time available is too short and hence all types of adjustments in the production process are impossible. As a plant capacity is fixed, output can be increased only by intensive utilization of existing plants and machinery or by having more shifts. Fixed factors remain the same and only variable factors can be changed to expand output. The total number of firms remains the same in the short period. Hence, total supply of the product can be adjusted to demand only to a limited extent.

In the short run, price is determined in the industry through the interaction of the forces of demand and supply. This price is given to the firm. Hence, the firm is a price taker and not a price maker. On the basis of this price, a firm adjusts its output depending on the cost conditions.

An industry under perfect competition in the short run reaches the position of equilibrium when the following conditions are fulfilled:

1. There is no scope for either expansion or contraction of the output of the entire industry. This is possible when all firms in the industry are producing an equilibrium level of output at which MR = MC. In brief, the total output remains constant in the short run at the equilibrium point. Thus, a firm in the short run has only **temporary equilibrium.**
2. There is no scope for the new firms to enter the industry or existing firms to leave the industry.
3. Short run demand should be equal to short run supply. The price so determined is called **'subnormal price'**. Normal price is determined only in the long run. Hence, short run price is not a stable price.

6. SHORT-RUN FIRM EQUILIBRIUM UNDER PERFECT COMPETITION

A competitive firm will reach an equilibrium position at a point where short run MR equals MC. At this point, equilibrium output and price are determined.

The firm in the short run will have only temporary equilibrium. The short run equilibrium price is not a stable price. It is also called a sub - normal price.

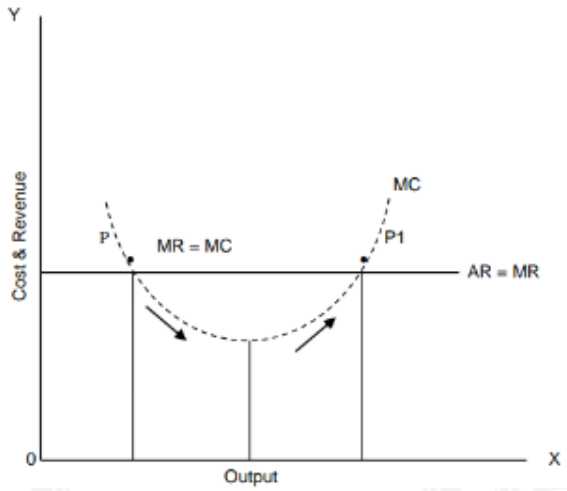


Figure 3: Short term equilibrium

The competitive firm, in the short run, will not be in a position to cover its fixed costs. But it must recover in the short run variable costs for its survival and continuance in the industry. A firm will not produce any output unless the price is at least equal to the minimum AVC. If the short run price is just equal to AVC, it will not cover fixed costs and hence, there will be losses. However, it will continue in the industry with the hope that it will recover the fixed costs in the future.

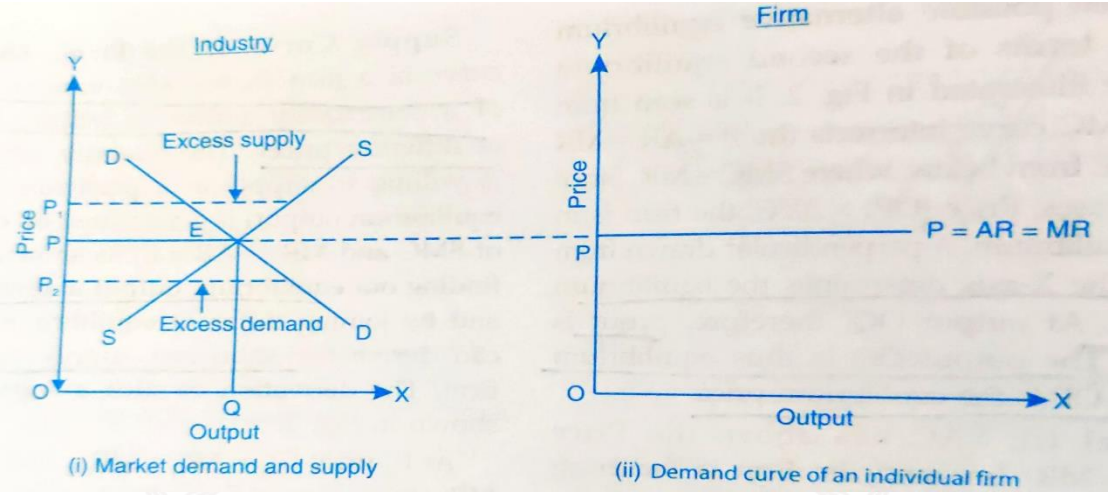


Figure 4: Determination of Market price and Demand Curve of an individual Firm

Source: Economics by DK Sethi.

A firm under perfect competition cannot influence the market price and is a price-taker. The firm is assumed to be facing a perfectly elastic demand for its product as any change in output over the range feasible for it will have a negligible effect on the total supply and the market price that the effect can safely assume to be Zero. The price OP is given to the firm. The perfectly elastic demand does not mean that the firm can sell an infinite amount at this given price, it only means that variation in the production feasible for it to make will leave price unaffected. This means the firm has not had to reduce the price to sell more. Also, it cannot sell anything if it charges more than the market price. Thus, a price taker firm always sells at the market price. Here Marginal revenue is equal average revenue at the level of market price at every level of output and the two would coincide in the same straight line. Thus AR = MR.

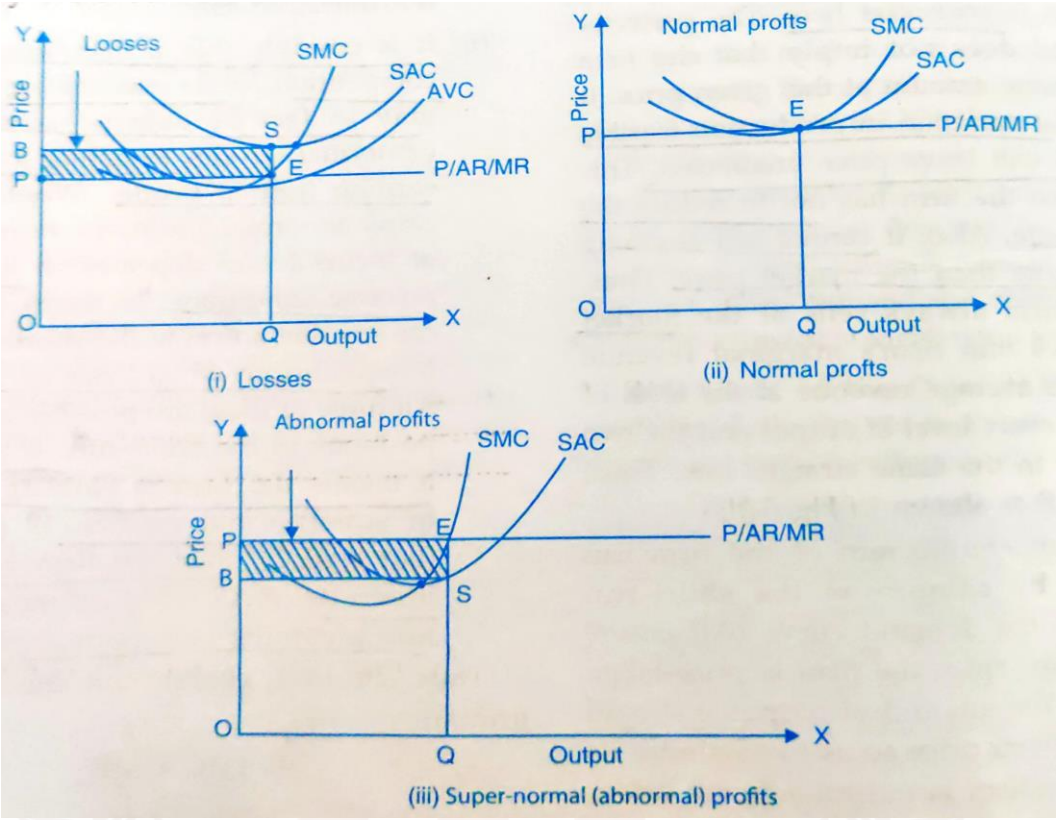


Figure 5: Short-run Equilibrium Position of a firm under Perfect Competition

In fig 5, In order to determine the profit maximizing output, the firm has to meet two types of conditions.

1. First condition for equilibrium at the firm in the short run is that it must ensure that SMC (Short run Marginal Cost) = MR, and SMC curve cuts the MR curve from below. Since for the perfectly competitive firm MR = AR = Price (P), it follows that this condition can be stated as SMC = P. This is the Profit maximization rule.
2. It is possible that a firm under perfect competition, while maximizing its profits, may in fact be earning super normal (abnormal) profits, incurring losses or just earning normal profits. All this depends on its cost and revenue conditions.

A firm in the short run will be in equilibrium, when price or AR intersects with SAC (short run Average Cost). All three of these are possible equilibrium positions in the short run.

a. If AR> SAC, it means the firm is earning abnormal or supernormal profits.
b. if AR < SAC (but > AVC), then the firm is incurring losses.
c. If AR = SAC, it means that the firm is earning only normal profits

In fig 5 it is seen that SMC curve intersects the P=AR=MR line at point E from below where SMC =MR since in all three cases price (OP) >AVC, the firm is in short run equilibrium. A Perpendicular drawn from point E to X-Axis determines the equilibrium output OQ. At Output OQ, therefore, profit is maximum. The output OQ is thus equilibrium output and OP is the equilibrium price.

In the panel(i), SAC lies above the price (P= AR=MR) line and the firm is incurring losses shown by the shaded area. Since prices exceed AVC, the firm continues to produce in the short run despite losses. Total losses will be equal to the area of the shaded rectangle. BSEP.

In Panel (ii), the SAC curve is tangent to the P=AR =MR line and the firm covers only its SAC, which includes normal profits. The firm therefore earns normal profits.

In the panel (iii)SAC curve lies below the P=AR=MR line at the equilibrium, and the firm earns super normal (abnormal) or pure profits, as shown by the shaded area BSEP.

7. LONG-RUN INDUSTRY EQUILIBRIUM UNDER PERFECT COMPETITION

In the long run, there is adequate time to make all kinds of changes, adjustments and readjustments in the production process. All factor inputs become variable in the long run. The total number of firms can be varied and plant capacity also can be changed depending upon the nature of requirements. Economies of scale, technological improvements, better management and organization may reduce production costs substantially in the long run. Hence, production can be either increased or decreased according to the needs of the individual firms and the industry as a whole. In short, the supply of the product can be fully adjusted to meet its demand in the long run period.

In the long run, an industry will reach the position of equilibrium under the following conditions:

1. At the point of equilibrium, the long run demand and supply of the products of the industry must be equal to each other. This will determine the long run normal price.
2. There will be no scope for the industry to either expand or contract output. Hence, total production remains stable in the long run.
3. All the firms in the industry should be in a position of equilibrium. All firms in the industry must be producing an equilibrium level of output at which long run MC is equated to long run MR. (MC = MR).
4. There should be no scope for entry of new firms into the industry or exit of old firms from the industry. In brief, the total number of firms in the industry should remain constant.
5. All firms should be earning only normal profits. This happens when all firms equate AR (Price) with AC. This will help the industry to attain a stable equilibrium in the long run.

8. LONG-RUN FIRM EQUILIBRIUM UNDER PERFECT COMPETITION

A competitive firm reaches the equilibrium position when it maximizes its profits. This is possible when:

1. The firm would produce that level of output at which MR = MC and MC curve cuts MR curve from below. The firm adjusts its output and the scale of its plant so as to equate MC with market price.

   Price = MC = MR

2. The firm in the long run must cover its full costs and should earn only normal profits. This is possible when the long-run normal price is equal to the long-run average cost of production. Hence,

   Price = AR = AC

3. When AR is greater than AC, supernormal profits are earned. This leads to the entry of new firms, an increase in the total number of firms, expansion in the output, an increase in the supply, a fall in the price and a fall in the ratio of profits. This process will continue till supernormal profits are reduced to zero. On the other hand, when AC is greater than AR, the industry will be incurring losses. This leads to the exit of old firms, decrease in the number of firms, contraction in output, rise in price, and rise in the ratio of profits. Thus, losses are avoided by automatic adjustments. Such adjustments will continue till the firm reaches the position of equilibrium when AC becomes equal to AR. Thus, losses and profits are incompatible with the position of equilibrium. Hence,

   Price = MR = MC = AR = AC

4. The firm is operating at its minimum AC making optimum use of available resources.

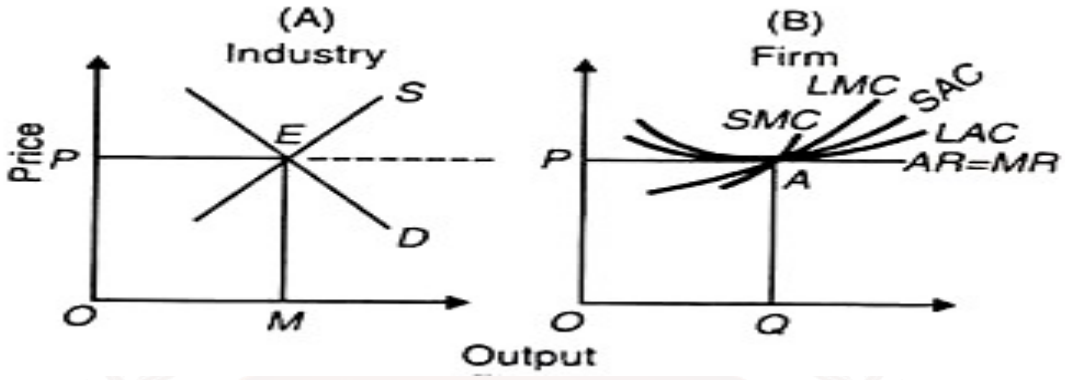


Figure 6: Long run equilibrium

In the case of the industry, E is the position of equilibrium at which LRS = LRD i.e. as per the fig 6 S is supply curve and D is demand curve intersect, indicating OP as the equilibrium price and OM as the equilibrium quantity demanded and supplied.

In the case of the firm, A indicates the position of equilibrium. At A, LMR = LMC and LMC curve cuts LMR curve from below. At the same point A, the minimum point of LAC is tangent to LAR curve. Hence,

LAR = LAC

A competitive firm in the long run must operate at the minimum point of the LAC curve. It cannot afford to operate at any other point on the LAC curve. Otherwise, it cannot produce the optimum output or it will incur losses.

Time plays an important role in determining the price of a product in the market. As the time under consideration is short, demand will have a more decisive role than supply in the determination of price. The longer the time under consideration, supply becomes more important than demand in the determination of price.

The price determined in the long run is called normal price and it remains stable.

Market price

Market price refers to that price which is determined by the forces of demand and supply in a short period where demand plays a major role and supply plays a passive role. Market price is unstable.

Normal price

Normal price is determined by demand and supply forces in the long period. It includes normal profits also and it is stable in nature.

SELF-ASSESSMENT QUESTIONS – 1

Fill in the blanks:	
1	The firms can earn only normal profit under _____ competition.
2	Under perfect competition, the demand curve is a _____ line.
3	Regardless of the cost structure of firms in a competitive market, in the long run, the marginal firm will earn_____ profit.
4	When new firms enter a perfectly competitive market the short-run market supply curve shift_____.
5	For any given price, a firm in a competitive market will maximize profit by selecting the level of output at which price intersects the_____ curve.
True or False Questions:	
6	In competitive markets, firms increasing their product prices earn larger profits. (True/False)
7	In perfectly competitive markets, firms are unable to differentiate their product from that of other producers. (True/False)
8	For a firm in a perfectly competitive market, marginal revenue is always equal to average revenue. (True/False)
9	A profit-maximizing firm in a perfectly competitive market will earn zero accounting profits in the long run. (True/False)
10	The marginal firm in a competitive market will earn zero economic profit in the long run. (True/False)

9. MONOPOLY

Meaning and definition

Monopoly means the existence of a single seller in the market. **Monopoly is that market form in which a single producer controls the whole supply of a single commodity which has no close substitutes.** Monopoly may be defined as a condition of production in which a single firm has the power to fix the price of the commodity or the output of the commodity. It is a situation where there exists a single control over the market producing a commodity having no substitutes with no possibilities for anyone to enter the industry to compete.

Features of monopoly

- ***Absence of competition*** **–** Absence of competition in the market creates a situation of monopoly and hence, the seller faces no threat of competition.
- ***Existence of a single seller*** **–** There will be only one seller in the market who exercises single control over the market.
- ***Absence of substitutes*** **–** There are no close substitutes for the seller's product with a strong cross elasticity of demand. Hence, buyers have no alternatives.
- ***Control over supply*** **–** Seller will have complete control over output and supply of the commodity.
- ***Price maker*** **–** The monopolist is the price maker, who is independent and can set the price to the best advantage. Hence, the monopolist can either charge a high price for all customers or adopt price discrimination policy if there are different types of buyers.
- ***Entry barriers*** **–** Entry of new firms is difficult. Hence, monopolists will not have direct competitors in the market.
- ***Firm and industry are the same*** **–** There will be no difference between the firm and an industry.
- ***Nature of firm*** **–** The monopoly firm may be a proprietary concern, partnership concern, Joint Stock Company or a public utility which pursues an independent price-output policy.
- ***Existence of super normal profits*** **–** There will be opportunities for supernormal profits under monopoly, because market price is greater than the cost of production.

There are different kinds of monopolies; private and public, pure monopoly, simple monopoly and discriminatory monopoly. It is to be clearly understood that with the exception of public utilities or

institutions of a similar nature, whose price is set by regulatory bodies, monopolies rarely exist. Just like perfect competition, pure monopoly also does not exist. Hence, we make a detailed study of simple monopoly and discriminatory monopoly in the foregoing analysis.

Price – output determination under monopoly Assumptions

- The monopoly firm aims at maximizing its total profit.
- It is completely free from government controls.
- It charges a single and uniform high price to all customers.

It is necessary to note that the price output analysis and equilibrium of the firm and industry is one and the same under monopoly.

As output and supply are under the effective control of the monopolist, the market forces of demand and supply do not work freely in the determination of equilibrium price and output in the case of the monopoly market. While fixing the price and output, the monopoly firm generally considers the following important aspects:

1. The monopolist can either fix the price of his/her product or its supply. The price and control of the supply cannot be fixed simultaneously. The price of the product may be fixed and supply can be determined by the demand conditions, or the output may be fixed and leaves the price to be determined by the demand conditions.
2. It would be more beneficial to the monopolist to fix the price of the product rather than fixing the supply, because it would be difficult to estimate the accurate demand and elasticity of demand for the products.
3. While determining the price, the monopolist has to consider the conditions of demand, cost of the product, possibility of the emergence of substitutes, potential competition, import possibilities, government control policies, etc.
4. If the demand for the product is inelastic, a relatively higher price can be charged and if the demand is elastic, a relatively lower price has to be charged.
5. Larger quantities can be sold at a lower price or smaller quantities at a higher price.
6. The most reasonable price should be charged, which is neither too high nor too low.
7. The most ideal price is that under which the total profit of the monopolist is the highest.

Price - output determination in the short run

Short run is a time period in which there are two types of factors of production. One is the fixed factors and the other is the variable factors. In the short period, production can be changed only by changing

the variable factors of production. In other words, in the short period, supply can be changed only to some extent, which is determined essentially by the capacity created. In this period, the volume of production can be changed but the capacity of the plant cannot be changed. The firm can increase the supply only with the help of existing machines and plants. New factories and plant- equipment cannot be installed.

The aim of a monopolist is to earn maximum profits or suffer minimum losses if the circumstances compel. Monopolist, being a single seller of the product, can fix the price equal to, above or less than the short period average cost of the product. Thus, normal profits or supernormal profits can be earned, or even incur losses in the short period.

The level of profit depends upon the nature and extent of the demand for the product. In order to earn maximum profits or suffer minimum losses, a monopolist compares the marginal revenue (MR) with marginal cost (MC). If marginal revenue exceeds marginal cost of a product, the monopolist can increase the profit by increasing production. On the contrary, if MC exceeds MR at a particular level of output, the monopolist can minimize the losses by reducing production. So, the monopolist is said to be in equilibrium, where marginal revenue is equal to marginal cost.

In the short period, a monopoly firm can earn supernormal profits, normal profits or incur losses. In case of losses, the price must cover at least the average variable costs. Otherwise, the firm will stop production. The maximum loss can be equal to fixed costs. Figure 1, 2 and 3 depict the three cases of monopoly equilibrium.

Three Cases of Monopoly Equilibrium

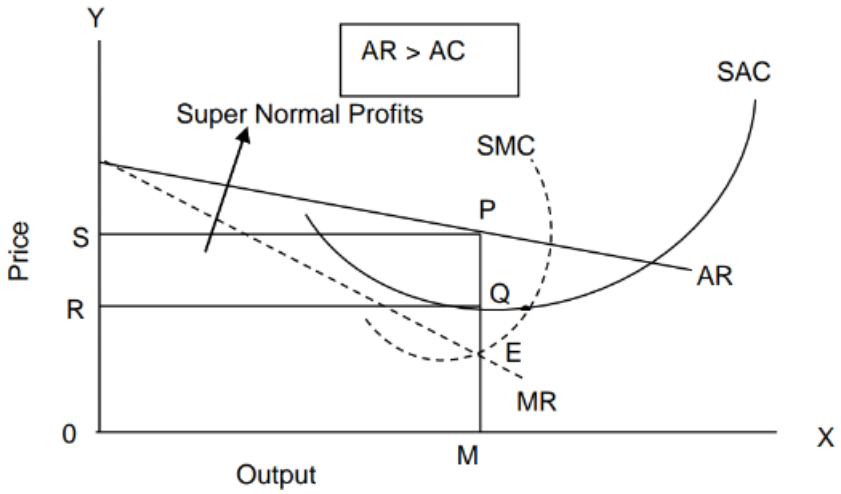


Figure 1: AR > AC

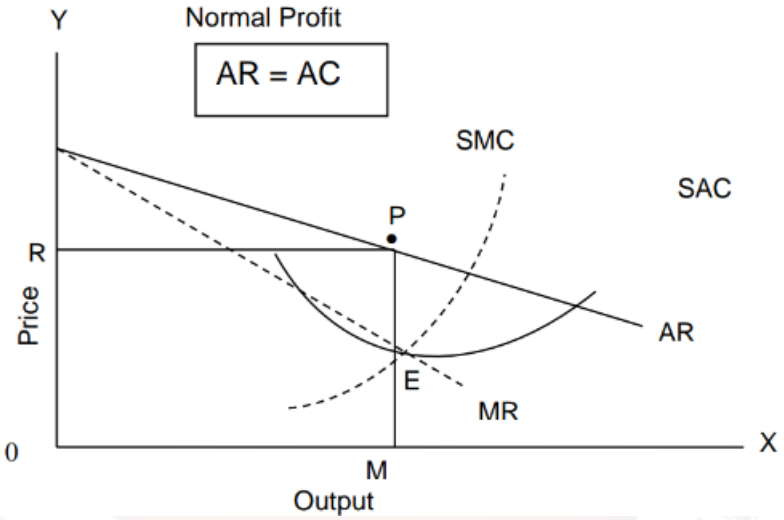


Figure 2: AR = AC

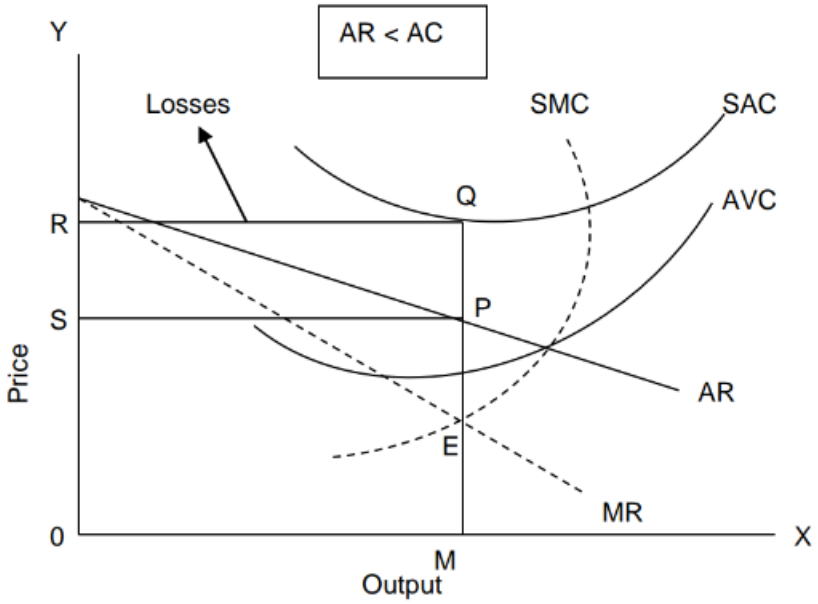


Figure 3: AR < AC

In figure 1: AR > AC. Hence, super normal profits.

In figure 2: AR = AC. Therefore, normal profits are incurred.

In figure 3: AR < AC. Therefore, losses are incurred.

The figures explain how a monopoly firm can earn supernormal profits, normal profits or incur losses in a short period.

Price – output determination in the long run

In the long run, there is adequate time to make all kinds of adjustments in both fixed as well as variable factor inputs. Supply can be adjusted to demand conditions. The total amount of long run profits will depend on the cost conditions under which the monopolist has to operate and the demand curve to be faced in the long run.

Under monopoly, the AR or demand curve drops downwards from left to right. This is because the monopolist can increase the sales and maximize the profits only when the price is reduced. MR is less than AR and hence, the MR curve lies below the AR curve. This is in accordance with the usual relationship between AR & MR.

The cost curve of the monopoly firm is influenced by the law of returns. The price to be charged for the product mainly depends on the nature of the cost curves.

The monopoly firm, in the long run, will continue its operations till it reaches the equilibrium point where long run MR equals long run MC. The price charged at this level of output is known as equilibrium price. In figure9.4, the monopoly firm reaches the position of equilibrium at E.

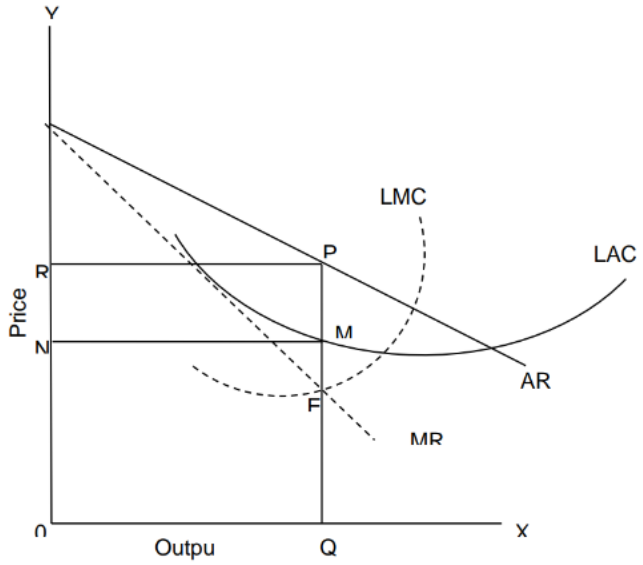


Figure 4: Long-run Price Determination in Monopoly

At this point, MR = MC and MC curve cuts MR curve from below. The monopolist will stop the output before AC reaches its minimum point and does not bother to reach the minimum point on AC.

The output is restricted in order to maximize the profit, OQ is the output. The price charged by the firm is QR (PQ), which is equal to AR. This price is higher than the average cost of QM per unit. The excess profit per unit of output is PM and the total profits of the firm is PM X RN = NRPM. Under monopoly, no doubt MR = MC but MR is less than AR**. Hence, under monopoly, price = AR only. Price is greater than AC, MC and MR.**

Generally speaking, monopoly price is slightly higher than that of competitive price because market price is over and above MC, MR and AC. The single seller has complete control over the supply to successfully prevent the entry of other new firms into the market. Thus, the monopoly power is reflected in its price. Monopoly prices are generally higher than competitive prices and thus detrimental to the interests of society.

Monopoly prices need not always be high on account of the following reasons:

- Due to the operation of both internal as well as external economies of scale, the monopolist may reduce the cost of production and hence, price too.
- The monopolist need not spend more money on sales promotion programmes. A lower price can be charged for the product and quite a lot of money can be saved.
- Monopolists fear that consumers may boycott the product if a high price is charged.
- There is fear of the discovery of new substitutes by other competitors in the market. Hence, low prices are charged.
- Fear of government intervention in controlling monopoly power leads to a lower price being charged.
- A lot of money may be spent on R&D and thus reduce the cost of operation. Cost reduction may facilitate price reduction.

Thus, in order to maintain the goodwill of the consumers and to secure good business, instead of charging high prices, the monopolist may charge a relatively lower price.

10. PRICE DISCRIMINATION UNDER MONOPOLY

Generally, the monopolist will not charge a uniform price for all the customers in the market. If it is beneficial to segment the market into homogenous groups, different methods will be followed under different circumstances. **The policy of price discrimination refers to the practice of a seller charging different prices for different customers for the same commodity, produced under a single control without corresponding differences in cost.** When a monopoly firm adopts this policy, it will become a discriminatory monopoly.4

Kinds of price discrimination:

Three kinds of price discrimination are commonly seen. It is as follows:

- **Discrimination of the first degree –** Under price discrimination of the first degree, the producer exploits the consumers to the maximum possible extent, by asking to pay the maximum price. In this case, the monopolist will not allow any consumer's surplus to the consumer. This type of price discrimination is called perfect discrimination.
- **Discrimination of the second degree –** In case of discrimination of the second degree, the monopolist charges different prices for markets of the same commodity, but not at a maximum possible rate but at a lower rate. The monopolist will leave a certain amount of consumer's surplus with the consumers. This is done to keep the consumers satisfied and prevent the entry of potential rivals. This method is adopted by railway companies.
- **Discrimination of the third degree –** In case of discrimination of the third degree, the markets are divided into many sub-markets or sub- groups. The price charged in each case roughly depends on the ability to pay of different subgroups in the market. This is the most common type of discrimination followed by a monopolist.

Price discrimination may take the following forms: (Basis of price discrimination)

1. **Personal differences**

   This refers to charging different prices for the same commodity due to personal differences arising out of ignorance and irrationality of consumers, preferences, prejudices and needs.

2. **Place**

   Markets may be divided on the basis of entry barriers, for example, the price of goods will be high in the place where taxes are imposed. Prices will be low in places where there are no taxes or low taxes.

3. **Different uses of the same commodity**

   When a particular commodity or service is meant for different purposes, different rates may be charged depending upon the nature of consumption. For example, different rates may be charged for the consumption of electricity for lighting, heating and productive purposes in industry and in agriculture.

4. **Time**

   Special concessions or rebates may be given during festival seasons or on important occasions.

5. **Distance**

   Railway companies and other transporters, for example, charge lower rates/km if the distance is long and higher rates if the distance is short.

6. **Special orders**

   When the goods are made to order, it is easy to charge different prices to different customers. In this case, a particular consumer will not know the price charged by the firm for other consumers.

7. **Nature of the product**

   Prices charged also depend on the nature of products, for example, railways charge higher prices for carrying coal and luxuries and lower prices for cotton, necessities of life, etc.

8. **Quantity of purchase**

   When customers buy large quantities, discounts will be allowed by the sellers. When small quantities are purchased, a discount may not be offered.

9. **Geographical area**

   Business enterprises may charge different prices at the national and international markets. For example, dumping – they charge a lower price in the competitive foreign market and a higher price in the protected home market.

10. **Discrimination on the basis of income and wealth**

    For example, a doctor may charge higher fees for rich patients and lower fees for poor patients.

11. **Special classification of consumers**

    For example, transport authorities such as railways and roadways show concessions to students and daily travelers. Different charges for I class and II class traveling, ordinary coach and air-conditioned coaches, special rooms and ordinary rooms in hotels, etc.

12. **Age**

    Cinema houses in rural areas and transport authorities charge different rates for adults and children.

13. **Preference or brands**

    Certain goods will be sold under different brand names or trademarks in order to attract customers. Different brands will be sold at different prices even though there is not much difference in terms of costs.

14. **Social and/or professional status of the buyer**

    A seller may charge a higher price for those customers who occupy higher positions and have a higher social status and a lower price for others.

15. **Urgency of the buyer**

    If a customer is in a hurry, a higher price would be charged. Otherwise, normal price would be charged.

16. **Discrimination on the basis of sex**

    In selling certain goods, producers may discriminate between male and female buyers by charging low prices to females.

17. **Peak season and off-peak season services**

    Hotel and transport authorities charge different rates during peak season and off-peak seasons. If price differences are minor, customers do not bother about such discrimination.

Pre-requisite conditions for price discrimination (When price discrimination is possible)

1. **Existence of imperfect market**

   Under monopoly, price discrimination occurs as there are buyers with incomplete knowledge and information about the market.

2. **Existence of different degrees of elasticity of demand in different markets**

   A monopolist will succeed in charging higher prices in the inelastic market and lower prices in the elastic market.

3. **Existence of different markets for the same commodity**

   This will facilitate price discrimination because buyers in one market will not know the prices charged for the same commodity in other markets.

4. **No contact among buyers**

   If there is a possibility of contact and communication among buyers, discriminatory practices followed by buyers will be known.

5. **No possibility of a resale**

   Monopoly products purchased by consumers in the low-priced market should not be resold in the high-priced market. Prevention of re exchange of goods is a must for price discrimination.

6. **Legal sanction**

   In some cases, price discrimination is legally allowed. For example, the electricity department will charge different rates per unit of electricity for different purposes. Similarly, charges on trunk calls, book post, registered posts, insured parcel, and courier parcel are different.

7. **Buyers' illusion**

   When consumers have an irrational attitude that high priced goods are of high quality, a monopolist can resort to price-discrimination.

8. **Ignorance and lethargy**

   Due to laziness and lethargy, consumers may not compare the price of the same product in different shops. Ignorance of consumers with regard to price variations would enable the monopolist to charge different prices.

9. **Preferences and Prejudices of buyers**

   The monopolist may charge different prices for different varieties or brands of the same product to different buyers. For example, low price for popular edition of the book and high price for deluxe edition.

10. Non-transferability features

In the case of direct personal services like private tuition, hair-cuts, beauty and medical treatments, a seller can conveniently charge different prices.

11. Purpose of service

The electricity department charges different rates per unit of electricity for different purposes like lighting, AEH, agriculture, industrial operations, etc. Railways charge different rates for carrying perishable goods, durable goods, necessaries and luxuries, etc.

12. Geographical distance and tariff barriers

When markets are separated by large distances and tariff barriers, the monopolist has to charge different prices due to the high transport cost, high rate of taxes, etc.

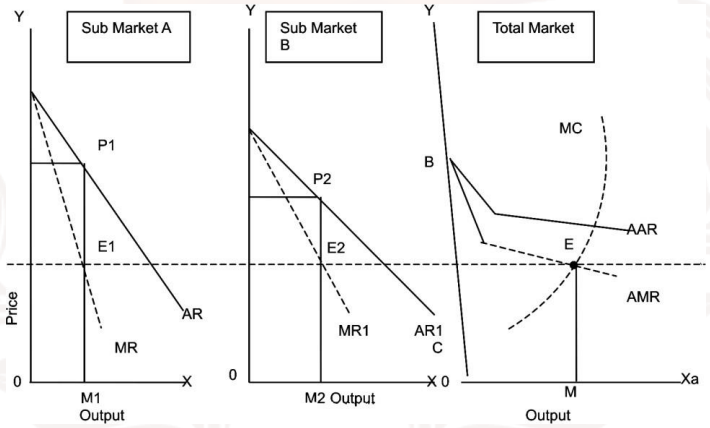


Figure 5: Pricing under discriminatory monopoly

Prices to be charged by the monopolist under price discrimination depend upon elasticity of demand for the products in different markets. The total output to be produced and supplied depends on marginal revenue and marginal cost. The principle of equilibrium under price discrimination is that marginal revenues in different markets are equal to marginal cost of the total output.

MR1 = MR2 = MC of the total output.

The monopolist, in order to exploit the market to the fullest extent, divides the market into two sub-markets, sub-market A and sub-market B, on the basis of price elasticity of demand. The total sales are distributed in these two markets. In sub-market A, the demand for the product is inelastic and hence a relatively higher price is charged for P1 & M1. The output sold in this market is OM1. E1 is the equilibrium position where MR = MC. P1 and E1 indicate the price over and above MR and MC.

In sub-market B, the demand for the product is relatively more elastic. A relatively lower price of P2 and M2 is charged. The output sold in this market is OM2. E2 is the equilibrium position where MR = MC. The distance between P2 & E2 indicates the excess of price over MR and MC.

The third diagram represents the total market for the product of the monopolist. AMR is the aggregate MR in both the markets and AAR is the aggregate AR in both the markets. MC is the marginal cost curve. At E, MR = MC, the equilibrium position of the monopoly firm. The total output sold in the two sub-markets is represented by OM.

The above description clearly shows that the monopolist has discriminated against the two markets and charged different prices in these two markets.

When price discrimination is profitable and justified:

- It is profitable for a monopolist when he or she charges a relatively higher price for those products having inelastic demand and a lower price for those having elastic demand. In this case, the total profits would certainly be high when compared to a simple monopolist who charges a single price.
- It is profitable to the general public only when price discrimination is allowed in public utility services and public sector where total receipts are lower than total costs. For example, railways, P & T, telephones, newspaper, housing, electricity department, water supply department, etc. Otherwise, common man and economically weaker section will not get certain products and services at cheaper rates. In such cases, from the point of view of their survival, growth and social welfare, price discrimination is justified.
- It is profitable when community's welfare requires price discrimination. For example, a doctor, or a lawyer, or a chartered accountant will charge a relatively higher fees for rich customers and lower fees for common man and poor people; this policy has redistributive effect. Inequalities in income and wealth distribution can be reduced through price discrimination. On the other hand, rich people can also get better service by paying higher fees or prices.
- It is profitable when the monopolist is organizing the production on a large-scale basis, increase the total output, reduce the production cost and charge a lower price in one market and a higher price in another market.
- It is profitable when the monopolist is sharing a part of the total profits with co-workers in the form of higher wages, salaries, bonuses, etc.

- It is profitable to the entire society when price discrimination leads to fuller utilization of national resources, higher output, income and employment and promotes the welfare of the general public.
- Normal dumping is profitable to a society, because surplus production is cleared off in foreign markets at lower prices and thus promotes export of the country.
- It is not profitable in cases where it leads to exploitation of the poor, common man, elimination of small entrepreneurs from the field of business, over investments in business, underutilization of resources and all other kinds of wastages, etc.

11. BILATERAL MONOPOLY

Bilateral monopoly is a special type of market situation in which a single seller faces a single buyer, i.e., a monopsonist is facing a monopolist. For example, in a town, there is only one steel factory offering employment for labor in the area and suppose a trade union controls the entire labor supply, the trade union which controls the supply of labor is the monopoly and the steel factory which is the sole buyer of labor is the monopsony.

In principle, the monopolist wishes to operate on a scale where the marginal cost is equal to marginal revenue, which will bring the maximum monopoly profit. On the other hand, the monopsonist wishes to purchase an amount at which marginal cost is equal to marginal utility. This indicates one optimum price for the buyer and another for the seller. There is, thus, indetermination in price fixation in bilateral monopoly. The two parties must enter into negotiations and the final price and quantity will depend upon the relative bargaining strength of the two parties. If the monopsonist is more powerful, the price will tend to be low; but if the monopolist is more powerful, the actual price will tend to be high.

Monopsony

Monopsony refers to a market with a single buyer who buys the entire amount produced. A Monopsony may be created when all the consumers of a commodity are organized together. For example, if there is only one cotton mill in a region, it becomes a monopsonist buyer of raw cotton, while the suppliers of cotton to the mill will be the large number of cotton growers. Just as the monopolist aims at maximizing profits, in the same manner, the monopsonist aims at maximizing consumer's surplus, and the consumer's surplus is maximum when the marginal cost is equal to marginal utility. A monopsonist too can adopt price discrimination, paying different prices to different sellers according to the elasticity of supply.

12. MONOPOLISTIC COMPETITION

Perfect competition and monopoly are two extreme forms of market situations, rarely to be found in the real world. Generally, markets are imperfect.

Prof. Chamberlin is the main architect of the theory of Monopolistic Competition. This market exhibits the characteristics of both perfect competition and monopoly. Since modern markets are combined and integrated with monopoly power and competitive forces, they are called Monopolistic Competition**. It is a market structure in which a large number of small sellers sell differentiated products which are close, but not perfect substitutes for one another.** Under this market, the products produced and sold are different, but they are close substitutes for one another. This leads to competition among different sellers. Thus, in this market situation, every producer is a sort of monopolist and between such "mini-monopolists" competition exists. It is one of the most popular and realistic market situations to be found in the present-day world. A number of examples may be given of this kind of market. Toothpaste, blades, motorcycles and bicycles, cigarettes, cosmetics, biscuits, soaps and detergents, shoes, ice – creams, suppliers of cane to a sugarcane factory in an isolated location where only one sugar factory exists, etc.

Characteristics of monopolistic competition

1. **Existence of a large number of firms**

   Under Monopolistic Competition, the number of firms producing a product will be large. The size of each firm is small. No individual firm can influence the market price. Hence, each firm will act independently without worrying about the policies followed by other firms. Each firm follows an independent price-output policy.

2. **The market is characterized by imperfections**

   Imperfections may arise due to advertisements, differences in transport cost, irrational preferences of consumers, ignorance about the availability of different brands of products and prices of products, etc., Sellers may also have inadequate knowledge about market and prices existing at different segments of markets.

3. **Free entry and exit of firms**

Each firm produces a very close substitute for the existing brands of a product. Thus, differentiation provides ample opportunity for a firm to enter with the group or an industry. On the contrary, if the firm faces the problem of product obsolescence, it may be forced to go out of the industry.

4. **Element of monopoly and competition**

Every firm enjoys some sort of monopoly power over the product it produces. However, it is neither absolute nor complete because each product faces competition from rival sellers selling different brands of the product.

5. **Similar products but not identical**

Under monopolistic competition, the firm produces commodities which are similar to one another but not identical or homogenous. For example, toothpastes, blades, cigarettes, shoes, etc.,

6. **Non-price competition**

In this market, there will be competition among "Mini-monopolists" for their products and not for the price of the product. Thus, there is "product competition" rather than "price competition".

7. **Definite preference of the consumers**

Consumers will have a definite preference for a particular variety or brand loyalty owing to the special features of a product produced by a particular firm.

8. **Product differentiation**

The most outstanding feature of monopolistic competition is product differentiation. Firms adopt different techniques to differentiate their products from one another. It may take mainly two forms:

a. **Real product difference:**

It will arise –

i. When products are produced from materials of higher quality, durability and strength.
ii. When products are extraordinary on the basis of workmanship, higher cost of material, color, design, size, shape, style, fragrance, etc.
iii. When personal care is taken to produce the products.

b. Imaginary product difference:

Producers adopt different methods to differentiate their products from that of other close substitutes in the following manner:

i. Proper location of sales depots in busy and prestigious commercial centers.
ii. Selling goods under different trademarks, patenting rights, different brands and packing them in attractive wrappers or containers.
iii. Providing convenient working hours to customers.
iv. Home delivery of goods with no extra cost.
v. Courteous treatment to customers, quick and prompt delivery of goods in time and developing cordial, personal and friendly relations with them.
vi. Offering gifts, discounts, lucky dip schemes, special prices, guarantee of repairs and other free services, guarantee of products, fair dealings, sales on credit or credit cards and debit cards, etc.
vii. Agreement to take back goods if they are unsatisfactory.
viii. Air-conditioned stores, etc.

9. Selling costs

All those expenses which are incurred on sales promotion of a product are called selling costs. In the words of Prof. Chamberlin – "selling costs are those which are incurred by the producers (sellers) to alter the position or shape of the demand curve for a product". In short, selling costs represent all those selling activities which are directed at persuading buyers to change their preferences so as to maximize the demand for a given commodity. Selling costs include expenses on sales depots, decoration of the shop, commission given to intermediaries, window displays, demonstrations, exhibitions, door to door canvassing, distribution of free samples, printing and distributing pamphlets, cinema slides, radio, T.V., newspaper advertisements (informative and manipulative advertisements), etc.

10. The concept of industry and product groups

Prof. Chamberlin introduced the concept of a group in place of industry. Industry in economics refers to a number of firms producing similar products. Under monopolistic competition, no doubt, different firms produce similar products but they are not identical. Hence, Prof. Chamberlin has made an attempt to redefine the industry. According to him, the monopolistically competitive

industry is a 'group 'of firms producing a 'closely related' commodity referred to as "product group". Thus, group refers to a collection of firms that produce closely related but not identical products.

11. More elastic demand curve

Product differentiation makes the demand curve of the firm much more elastic. It implies that a slight reduction in the price of one product, assuming the price of all other products remaining constant, leads to a large increase in the demand for the given product.

Price – output determination

short run equilibrium

A short period is a period of time where time is inadequate to make all sorts of changes and adjustments in the production process. The demand and cost conditions may vary substantially, forcing the firm either to charge a higher or lower price, leading to supernormal profits or losses. However, each firm fixes the price and produces output which maximizes its profit. The equilibrium price and output are determined at the point where short run marginal cost equals marginal revenue. Thus, the first condition for short run equilibrium is MC = MR in both diagrams.

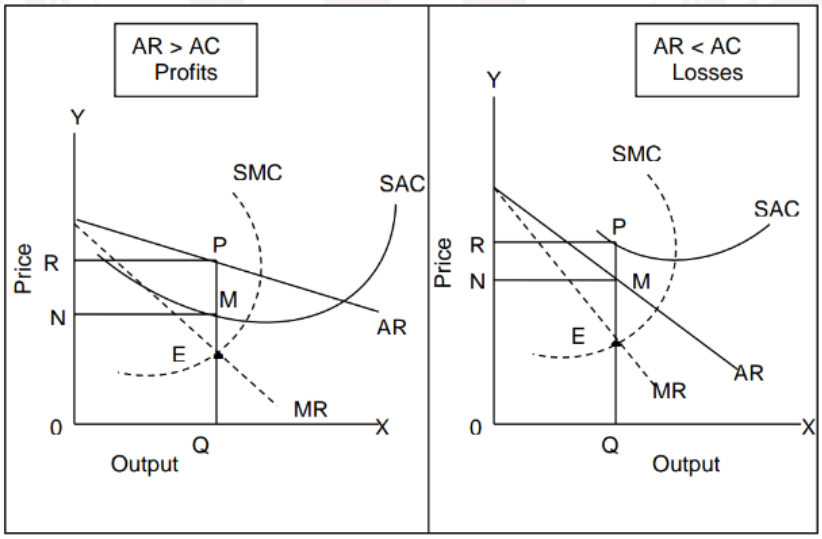


Figure 6: Short-run price determination in Monopolistic Competition

The first diagram shows supernormal profits. In this case, price (AR) is greater than AC (cost Per Unit). MQ is the cost per unit and total cost for OQ output is = MQ X OQ = ONMQ. PQ is the price or revenue per unit and the total revenue for OQ output is = PQ X OQ = ORPQ. Supernormal profit = TR (ORPQ) – TC (ONMQ). Hence, NRPM is the total profit.

The second diagram shows losses. In this case, AC is greater than AR. PQ is the cost per unit and the total cost is PQ x OQ = ORPQ. MQ is the revenue per unit and the total revenue for OQ output is MQ X OQ = ONMQ.

Total losses = TC (ORPQ) - TR (ONMQ) = NRPM. Thus, in the short run, there will be room for supernormal profits or losses.

Price output determination in the long run

Long run is a period of time where a firm will get adequate time to make any changes in the productive process or business. A firm can initiate several measures to minimize its production costs and enjoy all the benefits of large-scale production. The cost conditions, as a result, differ slightly in the long run. While fixing the price, a firm in the long run should consider its AC and AR.

Generally, in the long run, a firm can earn only normal profits. If AR is greater than AC, there will be supernormal profits. This leads to the entry of new firms – an increase in the total number of firms - total production – a fall in prices

- decline in profit ratio. On the other hand, if AC is greater than AR, there will be losses. This leads to the exit of old firms - decrease in the number of firms - total production - rise in prices – increase in profit ratio. Thus, the entry and exit of firms continue till AR becomes equal to AC. Thus, in the long run, two conditions are required for the equilibrium of the firm

1) MR = MC and

2) AR = AC. However, it should be noted that price is greater than MR and MC.

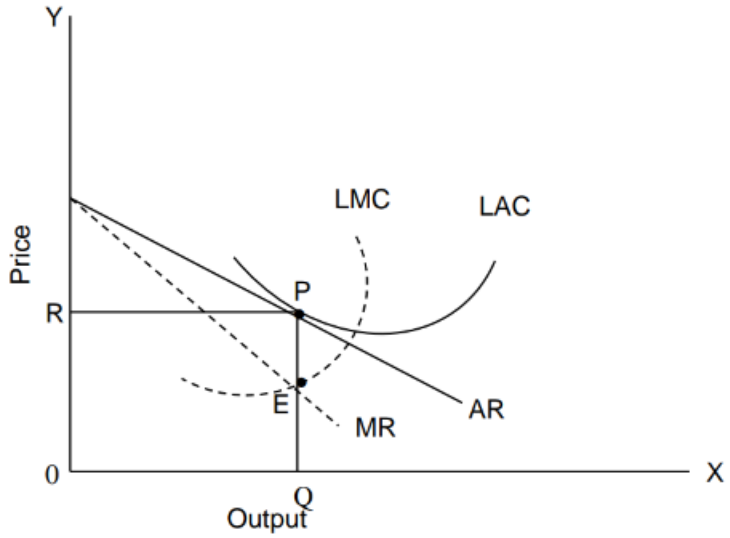


Figure 7: Long-run Price determination in Monopolistic Competition

In the diagram, E is the equilibrium position where MR = MC and MC curve cuts MR curve from below. At P, AR = AC = price.

It is necessary to understand that a firm under monopolistic competition in the long run can also earn supernormal normal profits. Prof. Stonier and Hague suggest that a firm can innovate, to introduce new changes in the context of a modern competitive business. This appears to be more realistic because today almost all firms make heavy profits. Hence, it is regarded as one of the most practical forms of market situations in the present day.

13. OLIGOPOLY

The term oligopoly is derived from two Greek words "Oligoi" means a few and 'Poly' means to sell. **Under oligopoly, we come across a few producers specializing in the production of identical goods or differentiated goods competing with one another.** The products traded by the oligopolists may be differentiated or homogeneous. In the case of the former, we can give the example of the automobile industry where different models of cars, ambassador, fiat, etc., are manufactured. Other examples are cigarettes, refrigerators, T.V. sets, etc. Pure or homogeneous oligopoly includes such industries as cooking and commercial gas, cement, food, vegetable oils, cable wires, dry batteries, petroleum, etc. In the modern industrial setup, there is a strong tendency towards oligopoly market situation. To avoid the impacts of competition in the case of competitive industries and to face the emergence of new substitutes in the case of monopoly industries, the oligopoly market is developed. For example, electric refrigerators, automatic washing machines, radio, etc.

Characteristics of oligopoly

1. **Interdependence**

   Each and every firm has to be conscious of the reactions of its rivals. Since the number of firms is very few, any change in price, output, product, etc. by one firm will have a direct effect on the policy of other firms. Therefore, economic calculations must be made always with reference to the reactions of the rival firms, as they have a high degree of cross elasticity of demand for their products.

2. **Indeterminateness of the demand curve**

   Under oligopoly, there will be an element of uncertainty. Firms will not be aware of the specific factors which could affect demand. Naturally, a rise or fall in the demand for the product cannot be speculated. The changes that would be taking place may be contrary to the expected changes in the product curve. Thus, the demand curve for the product will be indeterminate or indefinite. Prof. Sweezy explains it as a kinky demand curve.

3. **Conflicting attitude of firms**

   Under oligopoly, on one hand, firms may realize the disadvantages of competition and rivalry and desire to unite together to maximize their profits. On the other hand, firms guided by

individualistic considerations may continuously clashes with one another. This creates uncertainty in the market.

4. **Element of monopoly and competition**

   Under oligopoly, a firm has some monopoly power over the product it produces but not on the entire market. However, monopoly power enjoyed by the firm will be limited by the extent of competition.

5. **Price rigidity**

   Generally, prices tend to be sticky or rigid under oligopoly. This is because of the fact that if one firm changes its price, other firms may also resort to the same technique.

6. **Aggressive or defensive marketing methods -**

   Firms resort to aggressive and sometimes defensive marketing methods in order to either increase their share of the market or to prevent a decline of their share in the market. If one adopts extensive advertising and sales promotion policy it provokes others to do the same.

7. **Constant struggle**

   Competition is of a unique type in an oligopolistic market. Hence, competition consists of constant struggle of firms against rivals.

8. **Lack of uniformity**

   Lack of uniformity in the rise of different oligopolies is another remarkable feature.

9. **Small number of large firms**

   The number of firms in the market is small; but the size of each firm is big. The market share of each firm is sufficiently large to dominate the market.

10. **Existence of kinked demand curve**

    A kinked demand curve is said to occur when there is a sudden change in the slope of the demand curve. It explains price rigidity under oligopoly.

Price – output determination under oligopoly

There is no one system of pricing under the oligopoly market. Pricing policy followed by a firm depends on the nature of oligopoly and rivals' reactions. However, we can think of three popular types of pricing under oligopoly. They are as follows:

Independent pricing: (non-collusive oligopoly)

When goods produced by different oligopolists are more or less similar or homogeneous in nature, there will be a tendency for the firms to fix a common price. A firm generally accepts the "going price" and adjusts itself to this price. As long as the firm earns adequate profits at this price, it may not endeavor to change this price, as any effort to do so may create uncertainty. Hence, a firm follows what is called "acceptance pricing" in the market.

When goods produced by different firms are different in nature (differentiated oligopoly), each firm will be following an independent pricing policy as in the case of monopoly. In this case, each firm is aware of the fact that what it does would be closely watched by other oligopolists in the industry. However, due to product differentiation, each firm has some monopoly power. It is referred to as monopoly behavior of the Oligopolist. On the contrary, it may lead to price wars between different firms and each firm may fix price at a competitive level. A firm tends to charge prices even below its variable costs. This occurs as a result of one firm cutting the prices and others following the same. It is due to cut-throat competition in oligopoly. The actual price fixed by a firm may fall in between the upper limit laid down by the monopoly price and the lower limit fixed by the competitive price. It may be similar to that of the pricing under monopolistic competition. However, independent pricing in reality leads to antagonism, friction, rivalry, in-fighting, price-wars, etc., which may bring undesirable changes in the market. The oligopolist may realize the harmful effects of competition and may decide to avoid all kinds of waste. It encourages a tendency to come together. This leads to pricing under collusion. In other words, independent pricing can be followed only for a short period and it cannot last for a long period of time.

14. COLLUSIVE OLIGOPOLY AND PRICE LEADERSHIP

Pricing under collusion

Collusion is just the opposite of competition. The term collusion means to "play together" in economics. It means that the firms cooperate with each other in taking joint actions to keep their bargaining position stronger against the consumer. Firms give place for collusion when they join their hands in order to put an end to antagonism, uncertainty and its evils.

When government action is responsible for bringing the firms together, there will be room for explicit collusion. On the other hand, when restrictions are introduced, firms may form themselves into secret societies resulting in implicit collusion.

Collusion may be based on either oral or written agreements. Collusion based on oral agreement leads to the creation of what is called as "Gentlemen's Agreement ". It does not consist of any records. On the other hand, collusion based on written agreement creates what are known as cartels.

There are different types of cartel agreements. On one extreme, the firms surrender all their rights to a central authority which sets prices, determine output, marketing quotas for each firm, distributes profits, etc. This is called centralized cartels. A centralized or perfect cartel is an arrangement wherein the firms in an industry reach an agreement which maximizes joint profits. Hence, the cartel can act as a monopolist. Since the firms in the cartel are assumed to produce homogeneous goods, the market demand for the product is the cartel's demand. It is also assumed that the cartel management knows the demand at each possible price and also the marginal costs of all its firms. It can therefore find the MR and MC for the industry. The desire of the firms to have large joint profits gives an impulse to form cartels. But such a desire is short lived and therefore, the formal arrangement or cartels cannot be a long-term phenomenon.

Under the second type of cartel agreement, market – sharing cartel, the firms in the industry produce homogeneous products and agree upon the share each firm is going to have. Each firm sells at the same price but sells within a given region. Such a system can function only if the firms have identical costs.

The market sharing model has a very restrictive assumption of identical costs for all firms. Since, in practice, the firms have unequal costs and every firm wants to have some degree of independent action, the market-sharing cartels are not long-lived.

Price leadership

Perfect collusion is not possible in practice. Mutual suspicion and distrust among member-firms and their unwillingness to surrender all their Sovereignty makes collusion imperfect. There are a number of imperfect collusions and one of the most important ones is price leadership.

In this case, a particular strong firm which is enjoying the benefits of large-scale production will dominate the smaller firms. The price fixed by the dominant firm will be followed by all other small firms. Hence, the dominating firm becomes the price leader. All other firms following the price policy of the dominating firm in the industry are called price followers. The price leader is generally a leader in all markets. However, the same firm may become a follower in one market and a price leader in other markets. The leadership may emerge spontaneously due to technical reasons or out of tacit or explicit agreement between different firms to assign leadership roles to one of them. They may be either dominant-firm leadership or collusive-firm leadership.

Generally, leadership arises in a market for the following reasons:

• The leading firm will be enjoying the benefits of the lower cost of production and possess huge financial resources at its disposal.

- It may have a substantial share in the market.
- It will have a reputation for a sound pricing policy.
- It may take the initiative in dominating and controlling other firms in the industry as a normal method of functioning.
- It may follow an aggressive price policy and therefore it can acquire control over other firms.
- If a dominant firm is unable to perform its role as a leader either due to inherent deficiencies or government restrictions, it will assign the leadership role to other firms. This is called barometric price leadership.

The price leader has to make the following calculations before fixing the price of a given product:

- Reaction from rival firms for the product
- Elasticity of substitution between his/hers and others' product
- The price leader should follow an appropriate price policy whereby the leadership can be retained in the market. The support and loyalty of followers or price-takers should be taken. The guesswork of the leading firm while fixing the price should reflect the real condition in the

market. The price leader should be able to prevent other small firms from reducing the price to attract customers in the market.

- The price leader should remember that the power of the leader rests on the differences in costs. This type of price-leadership is called as partial monopoly.

Features of price leadership

- The price leader should be able to bear the risks of price wars in order to establish and maintain leadership. Once leadership is established, there should be persistent efforts to continue the lead.
- The price-leader normally takes the lead in increasing prices and in the case of price reductions; the leader becomes only a follower.
- Instead of thinking only about the short-term gains, the leader normally thinks about the long-term gains.
- The price leader has an important part in forecasting the demand, and cost condition in order to play his/her role effectively to win the confidence of the followers.
- Normally, the leader changes the price when changes in cost and demand conditions are permanent.
- The leader follows a definite and consistent pricing policy in a most intelligent manner so as to capture the market and to win over the small firms.
- An important aspect of price leadership is that it often serves as a means to price discipline and price stabilizations.

Advantages of price leadership: (for both leader and followers)

- Price leadership helps the small firms to formulate their price policy on the basis of the leader's price, as they do not normally possess complete information regarding various types of costs
- It is a simple and economical method of pricing because it does not involve any expenditure on market survey, etc.
- It ensures stability in the market by avoiding price wars as much as possible
- It will put an end to the operation of wide fluctuations in the market (operation of trade cycles)
- It reduces the number of reactions from different small firms and thus ensures certainty in the market
- It ensures the spirit of live and let live

Thus, price leadership has become an important method of pricing under oligopoly market conditions at present. It exists for a short period only. It is one of the most convenient methods of pricing for oligopoly firms which intend to stay and grow in the market.

Price - rigidity and kinked demand curve under oligopoly

The Kinked demand curve was first used by Prof. M. Sweezy to explain price rigidity under oligopoly. It represents the behavior of an oligopoly firm which has no incentive either to increase or decrease its price. Each firm, by its experience, has learnt what the reactions of rivals to actions will be on its part and may voluntarily avoid any activity that will lead to the situation of price war. Each firm is content with the present price output and profits and it does not want to make any change. Hence, they do not change their price- quantity combinations in response to small shifts in their cost curves.

When there are significant differences in quality, service and reputation in an industry, the price-leader operates in the upper quality stratum with a rich mixture of service, and charges some price premium for this superiority.

After a situation of price leadership is established, it is probably maintained fully as much by the followers as by the leader. The price-leader should make a temporary drop in price by informal concessions from the official price, because frequent changes in announced prices disrupt followers' adjustments and undermine the leader's prestige. Prices are changed only when changes in demand conditions and costs are permanent.

The price leader plays an important part in forecasting the demand and cost conditions to play the role effectively, accurately and in conformity with the confidence of followers.

The term 'Kink' refers to a short backward twist to cause obstructions. A kinked demand curve is said to occur when there is a sudden change in the slope of the demand curve. This gives rise to a kink, that is, a sharp corner in the demand curve. It arises when it is assumed that the rivals will lower their prices when the oligopolist lowers his/her own price; but the rivals will not raise their prices when the oligopolist raises his/her price.

Kinked demand curve analysis does not explain how price and output are determined under oligopoly rather, it seeks to explain why once a price-quantity combination has been established, a firm will avoid changing it, i.e. why the price becomes sticky. Hence, it provides an explanation to price rigidity under oligopoly conditions. In figure9.8, E represents the firm's original price-quantity combination.

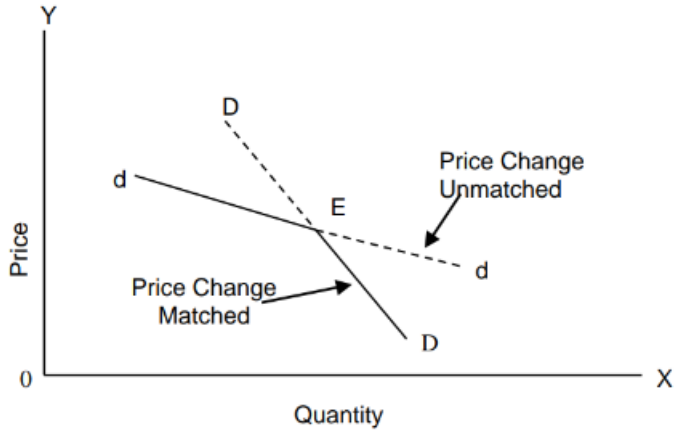


Figure 8: Firm's Original Price-Quantity Combination

When the oligopolist changes the price, the reaction of rivals will be as follows:

- Reaction to price reduction
- Reaction to price increase

Reaction to price reduction – If the oligopolist reduces the price while followers keep their price as constant, rival firms will experience a reduction in their demand and sales and a drift of customers to the oligopolist. So, they will also reduce their price to match the price reduction of the oligopolist. Even though the oligopolist reduces the price, there will not be any appreciable increase in demand for the product and also the sales. ED is the new demand curve which is inelastic. Even though prices fall, demand and sales will not go up considerably. Thus, this policy of price cut will not yield good results.

Reaction to price increase – When the oligopolist increases the price, the followers do not increase their prices. Now, rival firms get more customers because their prices are much lower than the oligopolist's price. Hence, without increasing their prices, the followers will earn more income. Now, the oligopolist, due to an increase in price, loses the demand and sales.

The demand curve will be the elastic segment DE.

An oligopolist faced with a kinked demand curve will be extremely unwilling to change the price; for a price decrease will cause no large increase in sales, and a price increase will cause a substantial fall in the sales. Thus, neither a price increase nor price reduction will be an attractive proposition for the oligopolist.

15. DUOPOLY

Features

Many economists are of the opinion that 'duopoly' is only a form of simple oligopoly. In other words, duopoly is only a limited oligopoly. **Duopoly is a market with two sellers exercising control over the supply of commodities.** It is a two-firm industry. Each seller knows that whatever he/she does will affect the rival's policies. Each seller attempts to make a correct guess of the rivals' motives and actions. The action by one will have a reaction from the other.

The two firms may either resort to competition or they may join together. On one extreme, the two rivals may go in for cut-throat competition with a view of eliminating the other from the market and setting themselves as monopolists. Such a type of competition may be disastrous for both. On the other hand, the rivals may realize that competition between them will ruin both and hence, they may fix the same price and restrict competition to advertising only.

Duopsony

Duopsony is an economic condition similar to a duopoly in which there are only two large buyers for a specific product or service. Members of a duopsony have great influence over sellers and can effectively lower market prices for their advantage.

For example, let's imagine a town in which only two restaurants operate. There are only two employment options for waiters and chefs. Since the restaurants have less competition for finding employees, they can offer lower wages. The chefs and waiters have no choice but to accept the low pay, unless they choose not to work. This shows that firms that are part of a duopsony have the power not only to lower the cost of supplies, but also to lower the price of labor.

Oligopsony

Similar to an oligopoly, this is a market in which there are a few large buyers for a product or service. This allows the buyers to have a great deal of control over the sellers and can effectively push down the prices.

16. INDUSTRY ANALYSIS

A detailed study of the market analysis enables us to understand how the business organization is influenced by the market structure, the conduct of the buyers and sellers and the overall performance of the market.

Structure

Under perfect competition, there are a large number of buyers and sellers, commodity dealt with are homogeneous and there is free entry and exit of firms into and out of the industry. Since the buyers and the sellers possess perfect knowledge of the market conditions, the same price prevails for the same commodity at the same time throughout the market. Such a situation promotes the welfare of both the buyers and the sellers and establishes ideal conditions of a market. It serves as a yardstick to measure the functioning of other markets.

Under monopoly, a single seller controls the entire market and has the power to control supply and price. Generally, a high price is charged by restricting the output. Product differentiation and selling costs dominate a monopolistic market. Intense competition prevails among the firms to promote the sale of their products. Oligopoly describes a situation where there are a few firms producing either homogeneous or differentiated products. Tough competition prevails among firms. Thus, different market structure explains different conditions under which a firm has to operate.

Conduct

The conduct of a firm depends upon the market structure in which it is operating. A firm under imperfect competition conducts more efficiently than under perfect competition. Under perfect competition, an individual firm has no control over price; it will have to adjust its output to the existing price in the market. Thus, the firm need not struggle much. Under imperfect competition, because of product differentiation and imperfect knowledge about the market, firms generally adopt aggressive sales promotion measures to survive and grow. Heavy investment is also made on research and development. Thus, there is incentive for growth and development. The welfare of the community is the highest.

Performance

Firms under perfect competition will have to perform efficiently to stay in the market. In the long run, price=MR=AR=MC=AC. As a rule, they cannot make abnormal profits. Under imperfect competition,

price is equal only to AR and AC in the long run; MC and MR are less than AR and AC. Thus, there is scope for supernormal profits. Under imperfect competition, firms perform better than the firms under perfect competition. Such an analysis of structure–conduct–performance of an industry is explained as structure- conduct-performance paradigm. Structure affects the conduct; conduct determines the performance. They are mutually interlinked.

SELF-ASSESSMENT QUESTIONS – 2

Fill in the blanks:	
1	_____ seller is the price maker and can restrict the output to increase the price.
2	_____ monopoly is a situation in which a single seller faces a single buyer.
3	Under _____ there are only two large buyers for a specific product or service.
4	_____ is the market situation where there is a monopoly element in case of the buyer.
5	_____ is a situation in which there are a few large buyers.
6	_____ costs are very important in a monopolistic market.
True or False Questions:	
7	A monopolist who suffers losses in the short run will exit in the long run if there is no plant size that will result in economic profit, that is greater than or equal to zero. (True/False)
8	If a monopolist is producing a level of output at which demand is inelastic, then the firm is not maximizing profit. (True/False)
9	If a monopolistically competitive market is in long-run equilibrium, each firm earns economic profits. (True/False)
10	Oligopolists face interdependent profits because industry sales are large. (True/False)

17. SUMMARY

Let us recapitulate the important concepts discussed in this unit:

- The organization and functioning of a firm are determined by the type of market in which it is operating.
- A market structure is characterized by the number of buyers and sellers, nature of the commodity dealt with, the scope for entry and exit of firms and the determination of price.
- Perfect competition exhibits an ideal market situation, where there are a large number of buyers and sellers, the commodity dealt with is homogeneous, there is free entry and exit of firms into and out of the industry, and a uniform price prevails in the market.
- In the long run Price is equal to MR=AR=MC=AC. The firms can make normal profit only in the long run.
- The organization and functioning of a firm is determined by the type of market in which it is operating.
- A market structure is characterized by the number of buyers and sellers, nature of the commodity dealt with, the scope for entry and exit of firms and the determination of price.
- Perfect competition exhibits an ideal market situation, where there are a large number of buyers and sellers, the commodity dealt with is homogeneous, there is a free entry and exit of firms into and out of the industry, and a uniform price prevails in the market. In the long run the price is equal to MR=AR=MC=AC. The firms can make only normal profit in the long run.
- Monopoly is a market situation where a single seller has total control over the price and output. There is scope for price discrimination. Different prices can be charged to different customers at different places; for different uses at different periods of time, for the commodity produced under the same cost conditions.
- Oligopoly is a market condition where a few big sellers producing either homogeneous or differentiated goods control the market. Popular methods of pricing under oligopoly are collusive pricing or price leadership.
- Bilateral monopoly is a situation where a monopolist faces a Monopsonist. Monopsony is a case of single buyers; Duopsony explains a situation of two buyers and Oligopsony, a situation of a few big buyers controlling the market.

- Industry analysis explains how the entire market is closely knit and how the structure of the market, conduct of the buyers and sellers, performance of the industry as a whole are all interrelated.

18. GLOSSARY

Firm	-	A single manufacturing unit producing and selling either a commodity or service.
Industry	-	All firms engaged in providing the same kind of service or doing common trade or business.
Market	-	A commodity and buyers and sellers of that commodity who are in competition with one another.
Market structure	-	Economically significant features of a market, which affect the behavior and working of firms in the industry.
Normal profits	-	Profits which are just sufficient to ensure the firms stay in business.
Perfectly competitive market	-	Market in which the number of buyers and sellers are large, all engaged in buying and selling a homogeneous product without any artificial restriction and possessing a perfect knowledge of the market at the same time.
Bilateral monopoly	-	A market situation in which a single seller faces a single buyer.
Dumping policy	-	This is the selling of goods at lower prices in the competitive international market and at higher prices in the protected domestic market.
Duopoly	-	A market with two sellers exercising control over the supply of commodities.
Duopsony	-	An economic condition in which there are only two large buyers for a specific product or service.
Monopolistic competition	-	A market structure in which a large number of small sellers sell differentiated products which are close, but not perfect substitutes for one another.
Monopoly	-	A market form in which a single producer controls the whole supply of a single commodity which has no close substitutes.

Monopsony	-	A market with a single buyer who buys the entire amount produced.
Oligopoly	-	A few producers specializing in the production of identical goods or differentiated goods competing with one another.
Price discrimination	-	The practices of a seller charging different prices for different customers for the same commodity, produced under a single control without corresponding differences in cost.

19. TERMINAL QUESTIONS

1. What are the primary characteristics of markets that categorize the market's structure?
2. Distinguish between a firm and an industry.
3. What is perfect competition? Explain how an industry's equilibrium price is determined under perfect competition in the short run.
4. When should a firm in a perfectly competitive market shut down its operation?
5. Explain the equilibrium of a firm under perfect competition in the long run.
6. What is monopoly?
7. What is price discrimination? Explain various kinds of price discrimination.
8. Define monopolistic competition and explain its characteristics.
9. What is oligopoly? Explain the features of oligopoly market.

20. ANSWERS

Self-Assessment Questions

1. Perfect
2. Horizontal
3. Zero Economic
4. Right
5. Marginal cost
6. False
7. True
8. True
9. False
10. True
11. Monopoly
12. Bilateral
13. Duopsony
14. Monopsony
15. Oligopsony
16. Selling
17. True
18. True
19. False
20. False

Terminal Questions Answers

Answer 1: The number and size distribution of sellers, the number and size distribution of buyers, product differentiation, conditions of entry and exit are all primary characteristics of market structure. Refer to section 2.

Answer 2: A firm is a single manufacturing unit producing and selling either a commodity or a service. Many firms producing similar or homogeneous goods or services collectively make an industry. Refer to section 4.

Answer 3: A perfectly competitive market is one in which the number of buyers and sellers are large, all engaged in buying and selling a homogeneous product without any artificial restriction and possessing perfect knowledge of the market at the same time. Refer to section 5.

Answer 4: At OP4 price, the firm will neither cover AFC nor AVC and hence it has to wind up its operations. It is regarded as a shut-down point. Refer to section 6.

Answer 5: A competitive firm reaches the equilibrium position when it maximizes its profits. Time plays an important role in determining the price of a product in the market. Refer to section 8.

Answer 6: Monopoly is that market form in which a single producer controls the whole supply of a single commodity which has no close substitutes. Refer to section Monopoly

Answer 7: The policy of price discrimination refers to the practice of a seller charging different prices for different customers for the same commodity, produced under a single control without corresponding differences in cost. Three kinds of price discrimination are Discrimination of the first degree, Discrimination of the second degree, Discrimination of the third degree. Refer to section on Price Determination

Answer 8: Monopolistic Competition is a market structure in which a large number of small sellers sell differentiated products which are close, but not perfect substitutes for one another. Its characteristics are definite preference of the consumers, market is characterized by imperfections, Existence of a large number of firms, free entry and exit of firms, Element of monopoly and competition, Product differentiation etc. Refer to section on Monopolistic Competition

Answer 9: Under oligopoly, we come across a few producers specializing in the production of identical goods or differentiated goods competing with one another. Its features are interdependence, indeterminateness of the demand curve, conflicting attitude of firms, lack of uniformity, constant struggle, aggressive or defensive marketing methods, price rigidity, small number of large firms etc. Refer to section on Oligopoly.

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